Experiment 5

Code:

```
import java.util.Scanner;


class Myexception extends Exception {
    public Myexception(String message) {
        super(message);
    }
}


class Bank {
    public void Bankdetails() {
        System.out.println("Bank Name: ICICI\nCity: Pune\nBank code: 112312");
    }
}


class Branch extends Bank {
    public void Branchdetails() {
        System.out.println("Branch_Id:151\nBranch Name:Kothrud
Branch\nAddress:Kothrud,Pune");
    }
}


class Account extends Bank {
    private String accountHolderName;
    private int accountNumber;
    private double balance;
```

```java
public Account(String accountHolderName, int accountNumber, double balance) {
    this.accountHolderName = accountHolderName;
    this.accountNumber = accountNumber;
    this.balance = balance;
}

public void Accountdetails() {
    System.out.println("Account Holder: " + accountHolderName);
    System.out.println("Account Number: " + accountNumber);
    System.out.println("Balance: " + balance);
}

public void deposit(double amount) {
    balance += amount;
    System.out.println("Deposited: " + amount);
}

public void withdraw(double amount) {
    balance -= amount;
    System.out.println("Withdrew: " + amount);
}

public void checkBalance() {
    System.out.println("Current Balance: " + balance);
}
```

```java
    public double getBalance() {
        return balance;
    }

    public int getAccountNumber() {
        return accountNumber;
    }
}

public class Exceptionhandling {

    static void valid(int accountNumber, double depositAmount) throws Myexception {
        if (accountNumber <= 0) {
            throw new Myexception("The account number is invalid");
        }
        if (depositAmount <= 0) {
            throw new Myexception("The amount to be deposited should be greater than 0");
        }
    }

    static void test(double balance, double amount) throws Myexception {
        if (amount > balance) {
            throw new Myexception("Your account balance is insufficient");
        }
        if (amount <= 0) {
            throw new Myexception("The amount to be withdrawn should be greater than 0");
        }
```

```
}

public static void main(String[] args) {
    final double initialBalance = 100000;
    Bank bank = new Bank();
    bank.Bankdetails();

    Branch branch = new Branch();
    branch.Branchdetails();

    Scanner scanner = new Scanner(System.in);
    try {
        System.out.println("\nEnter Account Holder Name:");
        String name = scanner.nextLine();

        System.out.println("Enter Account Number:");
        int accNum = scanner.nextInt();

        System.out.println("Enter amount to deposit:");
        double depositAmount = scanner.nextDouble();

        valid(accNum, depositAmount);

        Account account = new Account(name, accNum, initialBalance);
        account.deposit(depositAmount);

        System.out.println("\nAccount Details:");
```

```java
            account.Accountdetails();
            account.checkBalance();

            System.out.println("\nEnter amount to withdraw:");
            double withdrawAmount = scanner.nextDouble();

            test(account.getBalance(), withdrawAmount);
            account.withdraw(withdrawAmount);
            account.checkBalance();

        } catch (Myexception e1) {
            System.out.println("Exception: " + e1.getMessage());
        } finally {
            if (scanner != null) {
                scanner.close();
            }
        }
    }
}
```

Output:

```
Bank Name: ICICI
City: Pune
Bank code: 112312
Branch_Id:151
Branch Name:Kothrud Branch
Address:Kothrud,Pune

Enter Account Holder Name:
abcd
Enter Account Number:
100100
Enter amount to deposit:
500
Deposited: 500.0

Account Details:
Account Holder: abcd
Account Number: 100100
Balance: 100500.0
Current Balance: 100500.0

Enter amount to withdraw:
0
Exception: The amount to be withdrawn should be greater than 0
```